2026.10 | Refining Research Questions: OPDs and Computational Spectroscopy

2026-10-01 - 2026-10-31

1: Learning

- **Literature and research direction:** I continued studying dark current, charge collection, and bias-dependent response in OPDs. I also examined review topics for literature overlap, adequate controls, and usable data.

- **Computational spectroscopy:** Following discussions with my supervisor, I began focusing on single-pixel spectral reconstruction: how wavelength- and bias-dependent detector responses form a measurement matrix, and what spectral information that matrix can recover.

- **Mathematical foundations:** Using explanatory notes and a small reconstruction project, I studied least squares, the pseudoinverse, singular value decomposition (SVD), Tikhonov regularization, and nonnegative constraints, connecting these methods with noise amplification and nonunique solutions.

2: Highlights

- One initial digitized dataset contained twelve bias curves but about five numerical modes under an assumed 40 dB SVD threshold. This diagnostic calculation showed why more bias points need not provide more independent information; it was not a measured resolution result.

- I explored transfer-matrix and FDTD optics, drift-diffusion calculations, and CHARGE comparisons. I examined how thickness, contacts, and collection profiles affect spectral response, while treating experimental calibration as essential.

- I revised benchmarks to separate single-pixel devices from arrays and match wavelength ranges, test spectra, and error assumptions. Correcting a comparison weakened an apparent advantage over a published device and sharpened the question of what constituted a fair improvement.

- A back-to-back multilayer candidate led to a staged validation plan: measure calibration repeatability, examine doping and collection, test the difficult interface separately, and then consider the complete device. It remained a simulation proposal.

3: Insights & Takeaways

- Revising or stopping a direction became part of my research training. I set aside earlier review proposals when novelty checks and data coverage did not support their intended claims. Prior-art checks also helped distinguish a new implementation from an already established mechanism.

- For spectral reconstruction, a larger photocurrent is useful only in relation to the measurement task. The diversity of response shapes, their singular-value spectrum, calibration stability, and assumptions about the unknown spectrum all affect recoverable information.

- I learned to connect data analysis with physical interpretation by returning to the equations, examining model assumptions, and checking whether the results supported each conclusion.

4: Challenges & Open Questions

- The proposed device improvements were model-based. Estimated optical constants, transport parameters, and doping profiles limited their predictive strength; dark current, hysteresis, and nonlinear response were not fully represented in the candidate evaluation.

- Digitization and calibration errors could change the apparent number of usable modes. A numerical effective-rank threshold alone did not establish reconstruction accuracy or practical spectral resolution.

- I still needed deeper practice with regularization choices and independent checks of the forward model. Broad parallel exploration also made it important to prioritize a small number of questions with clear stopping criteria.

5: Following Plan

- Measure repeated multi-bias EQE spectra on existing devices to estimate calibration uncertainty, and check linearity, transient response, and illumination-history effects before interpreting reconstructed spectra.

- Test the candidate's most uncertain material and interface assumptions before attempting a complex stack, and update predictions with measured parameters.

- Continue targeted reading and consolidate one reproducible benchmark with explicit test spectra, error assumptions, matched comparisons, and criteria for continuing or revising the project.